

Cassandra Lochhaas

Institute for Theory and Computation Fellow
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[ADS Library Link](#)

EDUCATION

PhD in Astronomy <i>The Ohio State University</i>	July 2019 <i>Columbus, OH</i>
Master of Astronomy <i>The Ohio State University</i>	December 2015 <i>Columbus, OH</i>
Bachelor of Science in Physics <i>California Institute of Technology</i>	June 2013 <i>Pasadena, CA</i>

HIGHLIGHTS

- **Publications:** H-index 20, first-author papers typically cited 20-45 times, 9 first-author papers (225 total citations), 1 student first-author paper, 14 papers with substantial contributions (287 total citations), 9 other papers (626 total citations)
- **Grants, Fellowships, and Awards:** Harvard ITC Fellowship 2026-2028, NASA Hubble Fellowship 2023-2026 (\$397,000 over 3 years), PI of HST Theory grant (Cycle 28, \$137,400), AAS Rodger Doxsey Travel Prize (2019), Graduate Student Fellow and Presidential Fellow (OSU 2013 and 2018), 15 million CPU-hours on NASA supercomputer (\$200,000)
- **Talks:** 24 invited talks (including 8 international and 13 colloquia) and 4 contributed talks since 2019
- **Mentoring:** Sole scientific advisor for three and secondary mentor for two undergraduate research students, one student-led paper submitted to journal
- **Teaching/Outreach:** Ran The Ohio State University Planetarium for two years, gave 89 planetarium shows, developed one new show from scratch, and won the first installment of OSU's Ann S. Tuttle Citizenship, Engagement, and Outreach Prize
- **Leadership:** Won funding for and organized a CGM conference, organized FOGGIE group meetings (2020-present), organized two FOGGIE collaboration retreats, PI of next-generation FOGGIE simulations
- **Service:** Refereed 12 papers, reviewed applications for summer undergraduate research program, served as panel support for JWST review panel, served on NASA and NSF grants review panels, served on ITC fellowship selection committees

RESEARCH INTERESTS AND EXPERIENCE

Investigating link between galaxies and the circumgalactic medium (CGM) through developing and analyzing cosmological simulations and predicting CGM observations

Institute for Theory and Computation Fellow <i>Center for Astrophysics Harvard & Smithsonian</i> Studying how feedback drives turbulence and the impact of CGM turbulence on inflows	Sep 2026 - present <i>Cambridge, MA</i>
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NASA Hubble Fellow <i>Center for Astrophysics Harvard & Smithsonian</i> Building next-gen simulations, studying observability of the CGM and turbulence	Aug 2023 - Aug 2026 <i>Cambridge, MA</i>
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Postdoctoral Researcher*Space Telescope Science Institute*

Led projects exploring CGM structure in the FOGGIE group (PI Molly S. Peeples, Co-Is Jason Tumlinson & Brian W. O'Shea)

Sep 2019 - Aug 2023

*Baltimore, MD***PhD***The Ohio State University*

Thesis: Stellar feedback in galaxies, its impact on the circumgalactic medium, and the importance of radiative cooling. Advisors: Todd A. Thompson, David H. Weinberg, and Smita Mathur

Aug 2013 - Aug 2019

*Columbus, OH***Kavli Summer Program in Astrophysics Student Fellow***Center for Computational Astrophysics at the Flatiron Institute*

Led project evaluating turbulence in CGM simulations with Greg Bryan

June - August 2018

*New York, NY***TEACHING AND MENTORING EXPERIENCE**

- Mentor for Northeastern University Co-op Internship student** Jan 2025 - Aug 2026
Harvard & Smithsonian Center for Astrophysics *Cambridge, MA*
 - Developed a project and was the sole scientific advisor for undergrad Helena Bouchereau to measure velocity structure functions of turbulence in the CGM in cosmological zoom-in simulations
 - Mentored student through project with once to twice weekly meetings to teach advanced python, plot development and analysis, and the physics of turbulence
 - Coached writing of student-led paper (Bouchereau, **Lochhaas**, et al., submitted to ApJ Aug 2026)
 - Helena is beginning a physics master's program fall 2026
- Mentor for Maria Mitchell Observatory's REU Program** June - Aug 2024
Maria Mitchell Observatory *Nantucket, MA*
 - Developed a project and was the sole scientific advisor for undergrad Naija Bruckner to measure CGM turbulence using synthetic spectra of the CGM in cosmological zoom-in simulations
 - Mentored student through project with virtual meetings and a one week visit for intensive in-person mentoring, including background on the field, how to read in and analyze simulation data, produce and interpret scientific plots
 - Coached student through writing a report and giving a presentation on the main results
 - Naija is beginning a physics PhD program fall 2026
- Mentor for Space Astronomy Summer Program** June - Aug 2021
Space Telescope Science Institute *Baltimore, MD*
 - Developed a project and was the sole scientific advisor for undergrad Nami Nishimura to investigate the spread of CGM metallicity and inward and outward metal mass fluxes through the CGM using cosmological zoom-in simulations
 - Mentored student through project with minimum twice-weekly meetings, including background on the field, how to read in and analyze simulation data, produce and interpret scientific plots
 - Coached student through final presentation and interpreting main results
 - Nami is now in a physics graduate program
- Co-mentor for Space Astronomy Summer Program** June - Aug 2022, 2023
Space Telescope Science Institute *Baltimore, MD*
 - Was designated point-of-contact when primary mentor was unavailable
 - Helped two undergraduate students with interpretation and presentation of results over two summers

- **Planetarium TA** May 2015 - May 2017
The Ohio State University Columbus, OH
 - Scheduled planetarium shows, trained volunteers, produced software documentation and show scripts
 - Presented a total of 89 shows for the public, private field trip groups, and college classes
 - Attended the Spitz Summer Institute workshop in 2015 to learn how to produce effective planetarium shows
 - Developed new content, including an all-new full-length production, “Pluto: The Distant, Icy World”
- **TA for Introductory Astronomy courses** Spring Semester 2015
The Ohio State University Columbus, OH
 - Developed problems for homework sets, generated exams, quizzes, and study guides, graded all student work, held review sessions and office hours

SUCCESSFUL PROPOSALS AND GRANTS

- **NASA Hubble Fellowship** Aug 2023 - Aug 2026
Smithsonian Astrophysical Observatory Cambridge, MA
 “Non-Equilibrium Galaxy Evolution Driven by the CGM”
 \$397,000, 3 years, 7 million CPU-hours on NASA Pleiades computer (\$55,000)
 Faculty contact: Alexey Vikhlinin
- **Hubble Space Telescope Cycle 28 Archive Research Theory** Jan 2021 - Dec 2022
PI: Cassandra Lochhaas
 “What Holds Up the CGM?”
 \$137,400, 2 years, 8.75 million CPU-hours on NASA Pleiades computer (\$146,875)
 Co-Is: Molly S. Peeples, Jason Tumlinson, Brian W. O’Shea, Yong Zheng
- **Hubble Space Telescope Cycle 31 Archive Research Theory Grant** Jan 2024
PI: Anna Wright
 “Predicting Dwarf Galaxy Evolution in Resolved Milky Way Halos”
 Co-Is: Molly S. Peeples, Jason Tumlinson, Brian W. O’Shea, **Cassandra Lochhaas**, Raymond Simons, Erik Tollerud, Gregory Snyder
- **Hubble Space Telescope Cycle 28 Archive Research Theory Grant** Jan 2021
PI: Raymond Simons
 “On The Rapid Evolution of Galaxy Metallicity Gradients: A Bridge Between Theory and Observations”
 Co-Is: Molly S. Peeples, Jason Tumlinson, Brian W. O’Shea, Gregory Snyder, Ivelina Momcheva, **Cassandra Lochhaas**, Ramona Augustin, Casey Papovich, Jasleen Matharu

PENDING PROPOSALS

- **NASA Astrophysics Explorers Program 2026 Small Explorer (SMEX)** Sep 2026
PI: Sarah Tuttle
 “DISCO: Determining Intragalactic Structure with Cosmic Origins, A FUV two channel spectrograph to map cosmic ecosystems”
 Co-Is: Sanchayeeta Borthakur, Kate Rubin, Andrew Fox, **Cassandra Lochhaas**, Frances Cashman, Yakov Faerman
 My role: Vice-chair of GROOVE, one of the 3 surveys DISCO will accomplish, and leader of simulation comparisons
 My sub-award: \$160,000 over 4 years

- **MMT Observing Proposal** Oct 2026
*PIs: Sanchayeeta Borthakur and **Cassandra Lochhaas***
 “Mapping the Inner CGM of Galaxies with Binospec IFU: Connecting Feedback Processes to CGM Properties”
 Two proposals of 4 nights each through CfA and University of Arizona on MMT telescope using BINOSPEC instrument
 Co-Is: Brad Koplitz, Sebastian Lopez, Tanmay Singh, Yakov Faerman, Emma Kleiner, Namrata Roy, Sean Moran, Dennis Zaritsky

INVITED COLLOQUIA AND SEMINARS

- **Upcoming, “The Impact of Feedback and Circumgalactic Medium Turbulence on Galactic Accretion”** Dec 10, 2026
Astronomy Colloquium
U Mass Amherst, Amherst, MA
- **Upcoming, “The Impact of Feedback and Circumgalactic Medium Turbulence on Galactic Accretion”** Oct 28, 2026
Kapteyn Institute Astronomy Colloquium
University of Groningen, Groningen, Netherlands
- **“Galactic Accretion through the Dynamic Circumgalactic Medium”** Feb 3, 2025
Theoretical Astrophysics Center Seminar
UC Berkeley, Berkeley, CA
- **“The Non-Equilibrium Circumgalactic Medium”** May 21, 2024
Astronomy Colloquium
NRC Herzberg Centre, Victoria, Canada (virtual)
- **“The Non-Equilibrium Circumgalactic Medium”** Apr 4, 2024
Astronomy Seminar
UC Davis, Davis, CA (virtual)
- **“The Non-Equilibrium Circumgalactic Medium”** Dec 1, 2023
Astronomy Colloquium
New Mexico State University, Las Cruces, NM
- **“The Non-Equilibrium Circumgalactic Medium”** June 22, 2023
Diffuse Gas in Cosmic Ecosystems Special Interest Group
NASA Cosmic Origins Program Analysis Group (virtual)
- **“Circumgalactic Medium Dynamics are Crucial to Galaxy Evolution”** Feb 14, 2023
Astronomy Colloquium
University of Virginia, Charlottesville, VA
- **“Circumgalactic Medium Dynamics are Crucial to Galaxy Evolution”** Feb 1, 2023
Physics Colloquium
North Carolina State University, Raleigh, NC
- **“How Circumgalactic Medium Dynamics Impact Galaxy Evolution”** Nov 3, 2022
Astronomy Colloquium
University of Wisconsin Madison, Madison, WI
- **“Structure of the Circumgalactic Medium in High-Resolution Cosmological Simulations”** Dec 10, 2021
Astronomy Seminar
University of Pittsburgh/Carnegie Mellon University, Pittsburgh, PA (virtual)
- **“Structure of the Circumgalactic Medium in High-Resolution Cosmological Simulations”** Dec 8, 2021
Astronomy Seminar
Michigan State University, East Lansing, MI (virtual)

- **“Using Simulations to Understand the Structure of the Circumgalactic Medium”**
Astrophysics Seminar Oct 13, 2020
University of Notre Dame, Notre Dame, IN (virtual)

INVITED CONFERENCE TALKS

- **“The Impact of Feedback and Circumgalactic Medium Turbulence on Galactic Accretion”** Oct 9, 2026
Galaxies Across Cosmic Time Conference
Paso Robles, CA
- **“Simulated Mock Observations as a Powerful Tool for Probing Circumgalactic Medium Dynamics”** June 30, 2026
“Shaping Galaxies and their Gaseous Environments” Symposium
EAS 2026 Annual Meeting, Lausanne, Switzerland
- **“Simulation Predictions for the Metal Cycle with CGM Emission”** May 14, 2026
Galaxy Metal Cycles conference
Kylemore Abbey, Ireland
- **“Accretion through the Dynamic, Non-Equilibrium CGM”** Jan 8, 2025
Remembering Ed Jenkins: The Baryon Cycle
Princeton, New Jersey
- **“Accretion Streams through the CGM in the FOGGIE Simulations”** June 18, 2024
What Matter(s) Around Galaxies conference 2024
Varenna, Italy
- **“Accretion Streams through the CGM in the FOGGIE Simulations”** Mar 26, 2024
Arthur M. Wolfe Symposium 2024
UC San Diego, San Diego, CA
- **“The Non-equilibrium and Dynamic Circumgalactic Medium”** Sep 4, 2023
CGM at Notre Dame conference
Kylemore Abbey, Ireland
- **“How well do equilibrium theories characterize the CGM?” Invited Review** Feb 21, 2023
Oases in the Cosmic Desert conference
Arizona State University, Tempe, AZ
- **“Gas Kinematics Change the Temperature and Pressure of the CGM”** Dec 6, 2022
Aspera Science Workshop
University of Arizona, Tucson, AZ
- **“Tracking the Fuel for Galaxy Growth”** Nov 14-17, 2022
Supercomputing 2022 conference, NASA Exhibit
Dallas, TX
- **“How CGM Kinematics Affect More Than You Think”** Mar 22, 2022
Arthur M. Wolfe Symposium 2022
UC Santa Cruz, Santa Cruz, CA
- **“Impact of Galactic Winds on Circumgalactic Medium Structure”** Aug 20, 2021
Baltimore Winds Workshop
Johns Hopkins University, Baltimore, MD
- **“Thermal and Kinetic Properties of the Simulated Circumgalactic Medium”** June 24, 2020
The Circumgalactic Medium Around Galaxies Conference
Institut d’Astrophysique de Paris, Paris, France (virtual)

- **“Multiphase CGM: Fast Winds, Slow Shells”** Nov 7, 2019
Cosmic Turbulence and Magnetic Fields Conference
Institut d’Etudes Scientifiques de Cargèse, Corsica, France

CONTRIBUTED CONFERENCE TALKS

- **Led two discussion sessions on CGM turbulence** Aug 26 - Sep 6, 2024
Toward a Holistic Understanding of the Multi-scale, Multiphase Circumgalactic Medium
Aspen Center for Physics, Aspen, CO
- **“Properties of Accretion in the FOGGIE Simulations”** Aug 1, 2023
The Evolution of Gas in and Around Galaxies
Stanley, Idaho
- **“Outflow-driven CGM dynamics disrupt galactic accretion”** Jul 19, 2023
New Views on Feedback & the Baryon Cycle in Galaxies
Melbourne, Australia
- **“Impact of CGM Kinematics on Hot and Cold Mode Accretion”** Sep 15, 2022
What Matter(s) Around Galaxies conference 2022
Champoluc, Italy

AWARDS AND HONORS

- **American Astronomical Society Rodger Doxsey Travel Prize** Jan 2019
American Astronomical Society 233rd Meeting Seattle, WA
 - Travel funds awarded to 10 graduating PhD students to present dissertation work at AAS winter meeting
- **Ann S. Tuttle Citizenship, Engagement, and Outreach Prize** Dec 2018
The Ohio State University Department of Astronomy Columbus, OH
 - Awarded the first installment of this prize for exceptional outreach work
- **Presidential Fellow** Aug 2018 - Jul 2019
The Ohio State University Department of Astronomy Columbus, OH
 - Awarded to final-year graduate students for especially promising research in any field
- **Student Fellow, Kavli Summer Program in Astrophysics** June 2018 - Aug 2018
Center for Computational Astrophysics, Flatiron Institute New York, NY
 - 16 students invited to 6-week intensive summer program in galaxy evolution
- **Graduate Student Fellow** Sep 2013 - Aug 2014
The Ohio State University Columbus, OH
 - Awarded to especially promising incoming graduate students in any field
- **Robert L. Blinkenburg Summer Undergraduate Research Fellow** Summers 2011, 2012
California Institute of Technology Pasadena, CA
 - Summer research funding awarded to undergraduate researchers with interesting projects

LEADERSHIP

- Won funding for and organized a local conference: “Multiphase Madness: Resolving the CGM in Theory and Observations”, Center for Astrophysics, Aug 21-23, 2024
- Organized weekly FOGGIE research group meetings 2020-present, setting agenda and moderating discussion
- Organized biweekly development meetings for next-generation FOGGIE 2023-present: scheduled agenda items and led discussion

- Organized development of next-generation FOGGIE simulations 2023-present: Developed new code modules, developed analysis and testing framework, integrated others' code development, and led discussions and decisions about implementation and testing
- Helped organize and led several discussion sessions in three FOGGIE collaboration retreats, May 2024, March 2025, June 2026
- Organized new weekly research meeting at the Center for Astrophysics to discuss galaxy simulations and the circumgalactic medium with other postdocs and students, 2023-present

SERVICE

- Served on Harvard's Institute for Theory and Computation Fellowship admissions committees
- Refereed 12 papers
- Panel Support "Leveler" for JWST Cycle 1: listened to all panel discussions and ensured process remained fully anonymous
- Reviewed student applications for STScI Space Astronomy Summer Program, springs 2020-2022
- Served on NASA Astrophysics Theory Program Review Panels
- Served on NSF Astronomy & Astrophysics Grants Review Panel
- Organized the Institute for Theory and Computation Discussion series, Center for Astrophysics, 2023-2024

REFERENCES

- Dr. Jason Tumlinson, Space Telescope Science Institute. tumlinson@stsci.edu
- Prof. Sanchayeeta Borthakur, Arizona State University. sborthak@asu.edu
- Prof. Charlie Conroy, Harvard University. cconroy@cfa.harvard.edu

Cassandra Lochhaas - Publication List

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ADS Library: ui.adsabs.harvard.edu/public-libraries/IbvAMyU1SaihKteJvJv_HA

FIRST AUTHOR PUBLICATIONS

1. **Cassandra Lochhaas**, Jason Tumlinson, Molly S. Peeples, Brian W. O’Shea, et al., “Figuring Out Gas & Galaxies In Enzo (FOGGIE) XI: Circumgalactic O VI Emission Traces Clumpy Inflowing Recycled Gas”, 2026, *ApJ*, 1000, 104
2. **Cassandra Lochhaas**, Jason Tumlinson, Molly S. Peeples, Brian W. O’Shea, et al., “Figuring Out Gas & Galaxies In Enzo (FOGGIE) VI: The Circumgalactic Medium of L^* Galaxies is Supported in an Emergent, Non-Hydrostatic Equilibrium”, 2023, *ApJ*, 948, 43
3. **Cassandra Lochhaas**, Jason Tumlinson, Brian W. O’Shea, Molly S. Peeples, et al., “Figuring Out Gas & Galaxies In Enzo (FOGGIE) V: The Virial Temperature Does Not Describe Gas in a Virialized Galaxy Halo”, 2021, *ApJ*, 922, 121
4. **Cassandra Lochhaas**, Todd A. Thompson, & Evan E. Schneider, “The Characteristic Momentum of Radiatively Cooling Energy-Driven Galactic Winds”, 2021, *MNRAS*, 504, 3412
5. **Cassandra Lochhaas**, Greg L. Bryan, Yuan Li, et al., “Properties of the Simulated Circumgalactic Medium.” 2020, *MNRAS* 493, 1461
6. **Cassandra Lochhaas**, Smita Mathur, Stephan Frank, et al., “A High Signal-to-Noise HST Spectrum Toward J1009+0713: Precise Absorption Measurements and the Origin of O VI.” 2019, *MNRAS* 489, 78
7. **Cassandra Lochhaas**, Todd A. Thompson, Eliot Quataert, et al., “Fast Winds Drive Slow Shells: A Model for the CGM as Galactic Wind-Driven Shells.” 2018, *MNRAS*, 481, 1873
8. **Cassandra Lochhaas** & Todd A. Thompson, “Second Generation Stars in Globular Clusters from Rapid Radiative Cooling of Pre-Supernova Massive Star Winds.” 2017, *MNRAS*, 470, 977
9. **Cassandra Lochhaas**, David H. Weinberg, Sébastien Peirani, et al. “Modeling Lyman- α Forest Cross-Correlations with LyMAS.” 2016, *MNRAS*, 461, 4353

STUDENT PUBLICATIONS

1. Helena Bouchereau, **Cassandra Lochhaas**, Jake S. Bennett, & Mandy C. Chen, “Measuring Simulated Circumgalactic Medium Turbulence with Emission-Weighted Projected Velocity Structure Functions in FOGGIE”, 2026, *submitted to ApJ*, arXiv:2608.17013

SIGNIFICANT CONTRIBUTION PUBLICATIONS

1. Mackenzie Tcoras, Brian W. O’Shea, Claire Kopenhafer, et al. (**Cassandra Lochhaas** 4th author of 12), “Figuring Out Gas & Galaxies in Enzo (FOGGIE). XV. Examining the Spatial and Kinematic Relationship between Circumgalactic Mg II and O VI”, 2026, *ApJ*, 1004, 159
2. Vida Saeedzadeh, Jason Tumlinson, Molly S. Peeples, et al. (**Cassandra Lochhaas** 5th author of 9), “Figuring Out Gas & Galaxies In Enzo (FOGGIE). XIV. The Observability of Emission from Accretion and Feedback in the Circumgalactic Medium with Current and Future Instruments”, 2026, *ApJ* 1004, 9
3. Cameron W. Trapp, Molly S. Peeples, Jason Tumlinson, et al. (**Cassandra Lochhaas** 5th author of 11), “Figuring Out Gas & Galaxies In Enzo (FOGGIE). XII. The Formation and Evolution of Extended HI Galactic Disks and Warps with a Dynamic CGM”, 2026, *ApJ* 1003, 186

4. Emily Silich, John ZuHone, Elena Bellomi, et al. (**Cassandra Lochhaas** 7th author of 9), “X-ray emission signatures of galactic feedback in the hot circumgalactic medium: predictions from cosmological hydrodynamical simulations”, 2025, *ApJ* 993, 125
5. Emily M. Witt, Brian T. Fleming, Stephan McCandliss, et al. (**Cassandra Lochhaas** 8th author of 10), “The Juniper CubeSat concept: mission overview”, 2025, *Proc. SPIE* 13625, 136250L
6. Ramona Augustin, Jason Tumlinson, Molly S. Peeples, et al. (**Cassandra Lochhaas** 6th author of 14), “FOGGIE X: Characterizing the Small-Scale Structure of the CGM and its Imprint on Observables”, 2025, *ApJ* 993, 52
7. Raymond C. Simons, Molly S. Peeples, Jason Tumlinson, et al. (**Cassandra Lochhaas** 5th author of 13), “Figuring Out Gas & Galaxies in Enzo (FOGGIE). IX: The Angular Momentum Evolution of Milky Way-like Galaxies and their Circumgalactic Gas”, 2025, *ApJ* 988, 250
8. Alison L. Coil, Serena Perrotta, David S. N. Rupke, et al. (**Cassandra Lochhaas** 4th author of 12), “Detection of Spatially Extended Ionized Gas in an Odd Radio Circle”, 2024, *Nature* 625, 459
9. Ayan Acharyya, Molly S. Peeples, Jason Tumlinson, et al. (**Cassandra Lochhaas** 5th author of 10), “Figuring Out Gas & Galaxies In Enzo (FOGGIE) VIII: Complex and Stochastic Metallicity Gradients at $z > 2$ ”, 2025, *ApJ* 979, 129
10. Hyeonmin Lee, Ayan Acharyya, Anna C. Wright, et al. (**Cassandra Lochhaas** 5th author of 7), “Evolution of Galaxy Size in the FOGGIE Simulations”, 2023, *Research Notes of the AAS*, 7, 202
11. Anna C. Wright, Jason Tumlinson, Molly S. Peeples, et al. (**Cassandra Lochhaas** 5th author of 10), “Figuring Out Gas & Galaxies In Enzo (FOGGIE) VII: The (Dis)Assembly of Stellar Halos”, 2024, *ApJ* 970, 70
12. Raymond C. Simons, Molly S. Peeples, Jason Tumlinson, et al. (**Cassandra Lochhaas** 7th author of 12), “Figuring Out Gas & Galaxies In Enzo (FOGGIE). IV. The Stochasticity of Ram Pressure Stripping in Galactic Halos”, 2020, *ApJ* 905, 167
13. Yong Zheng, Molly S. Peeples, Brian W. O’Shea, et al. (**Cassandra Lochhaas** 5th author of 9), “Figuring Out Gas & Galaxies In Enzo (FOGGIE). III. The Mocky Way: Investigating Biases in Observing the Milky Way’s Circumgalactic Medium”, 2020, *ApJ* 896, 143
14. Yuan Li, Marie-Lou Gendron-Marsolais, Irina Zhuravleva, et al. (**Cassandra Lochhaas** 7th author of 19), “Direct Detection of Black Hole-driven Turbulence in the Centers of Galaxy Clusters”, 2020, *ApJL* 889, 1

OTHER PUBLICATIONS

1. Jake S. Bennett, Aaron Smith, Fabrizio Arrigoni-Battaia, et al. (**Cassandra Lochhaas** 5th author of 6), “The birth of the intracluster medium: the evolution of multiphase gas and Lyman- α haloes in a simulated $z \sim 3$ protocluster”, 2026, *OJAp*, 9, 65188
2. Alison L. Coil, David S. N. Rupke, Serena Perrotta, et al. (**Cassandra Lochhaas** 5th author of 5), “A Shocked Wind Interpretation of an Odd Radio Circle”, 2026, *ApJ*, 1003, 24
3. Keith Horne, G. De Rosa, B. M. Peterson, et al., “Space Telescope and Optical Reverberation Mapping Project. IX. Velocity-Delay Maps for Broad Emission Lines in NGC 5548”, 2021, *ApJ* 907, 76
4. Williams, P. R., Pancoast, A., Treu, T., et al., “Space Telescope and Optical Reverberation Mapping Project. XII. Broad-Line Region Modeling of NGC 5548”, 2020, *ApJ* 902, 74
5. G. A. Kriss, G. De Rosa, J. Ely, et al., “Space Telescope and Optical Reverberation Mapping Project. VIII. Time Variability of Emission and Absorption in NGC 5548 Based on Modeling the Ultraviolet Spectrum”, 2019, *ApJ* 881, 153

6. G. De Rosa, M. M. Fausnaugh, C. J. Grier, et al., “Velocity-resolved Reverberation Mapping of Five Bright Seyfert 1 Galaxies.” 2018, ApJ 866, 133
7. M. M. Fausnaugh, D. A. Starkey, Keith Horne, et al., “Continuum Reverberation Mapping of the Accretion Disks in Two Seyfert 1 Galaxies.” 2018, ApJ, 854, 107
8. M. M. Fausnaugh, C. J. Grier, M. C. Bentz, et al., “Reverberation Mapping of Optical Emission Lines in Five Active Galaxies.” 2017, ApJ, 840, 97
9. L. Pei, M. M. Fausnaugh, A. J. Barth, et al., “Space Telescope and Optical Reverberation Mapping Project. V. Optical Spectroscopic Campaign and Emission-line Analysis for NGC 5548.” 2017, ApJ, 837, 131